WORKED SOLUTION

STEP III 2009, Question 13

Probability and Statistics • Central Limit Theorem

STEP III 2009, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/09-s3-q13

The key idea

Everything is a function of the single angle θ: $X = \cos\theta$ and $Y = \sin\theta$. So the distribution of X comes from the cumulative distribution function (work out which arc of the circle gives $\cos\theta \le x$), and the means, variances and covariances are just averages of trigonometric functions over a full turn. In part (ii), $\overline{X}$ is an average of many independent, identically distributed variables, which is exactly the situation where the central limit theorem applies.

Solution

Part (i)

Since P is on the unit circle, $X = \cos\theta$ and $Y = \sin\theta$, where θ has density $\frac{1}{2\pi}$ on $[0, 2\pi)$.

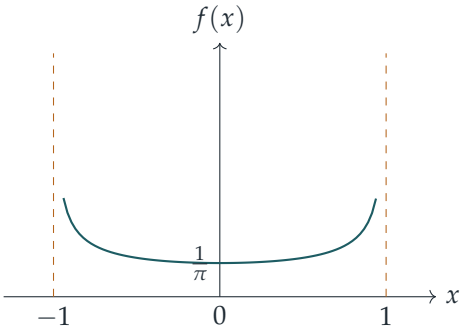
The density of X. Clearly $-1 \le X \le 1$. Take $-1 < x < 1$ and let $\alpha = \arccos x \in (0, \pi)$. On $[0, 2\pi)$ we have $\cos\theta > x$ exactly when $\theta < \alpha$ or $\theta > 2\pi - \alpha$, an arc of total length 2α. Hence

$$F(x) = \mathrm{P}(X \le x) = 1 - \frac{2\alpha}{2\pi} = 1 - \frac{\arccos x}{\pi}.$$

Differentiating, using $\dfrac{\mathrm{d}}{\mathrm{d}x}\arccos x = -\dfrac{1}{\sqrt{1-x^2}}$,

$$f(x) = \frac{1}{\pi\sqrt{1-x^2}} \qquad (-1 < x < 1),$$

and $f(x) = 0$ otherwise. (A quick check: $\int_{-1}^{1} f(x)\,\mathrm{d}x = \frac{1}{\pi}[\arcsin x]_{-1}^{1} = 1$.)



The density is smallest at $x = 0$ and blows up at $x = \pm 1$: the point P moves horizontally slowest near the ends of the horizontal diameter, so X spends most of its time there.

The variance. By symmetry (f is even), $\mathrm{E}(X) = 0$. Working directly with θ,

$$\mathrm{E}(X^2) = \frac{1}{2\pi}\int_0^{2\pi} \cos^2\theta \,\mathrm{d}\theta = \frac{1}{2\pi}\int_0^{2\pi} \tfrac{1}{2}(1+\cos 2\theta)\,\mathrm{d}\theta = \frac{1}{2},$$

so

$$\mathrm{Var}(X) = \mathrm{E}(X^2) - (\mathrm{E}(X))^2 = \tfrac{1}{2}.$$

By the same argument (or since $\theta - \frac{\pi}{2}$ is also uniform on a full turn), $\mathrm{E}(Y) = 0$ and $\mathrm{Var}(Y) = \frac{1}{2}$.

Uncorrelated. We have

$$\mathrm{E}(XY) = \frac{1}{2\pi}\int_0^{2\pi} \cos\theta \sin\theta \,\mathrm{d}\theta = \frac{1}{4\pi}\int_0^{2\pi} \sin 2\theta \,\mathrm{d}\theta = 0,$$

so $\mathrm{Cov}(X, Y) = \mathrm{E}(XY) - \mathrm{E}(X)\mathrm{E}(Y) = 0 - 0 = 0$. Hence X and Y are uncorrelated.

Not independent. They are certainly not independent, since $X^2 + Y^2 = 1$: once X is known, Y is pinned down to one of $\pm\sqrt{1 - X^2}$. For a concrete event, $\mathrm{P}(X > 0.8) > 0$ and $\mathrm{P}(Y > 0.8) > 0$, but $\mathrm{P}(X > 0.8 \text{ and } Y > 0.8) = 0$ because then $X^2 + Y^2 > 1.28$. Independence would require this joint probability to equal the (positive) product. Zero correlation only says there is no *linear* relationship; here the relationship is a circle, which is symmetric enough to have no linear trend at all.

Part (ii)

Each pair (X_i, Y_i) has the distribution in part (i), and pairs with different indices are independent because the points are chosen independently.

$\overline{X}$ and $\overline{Y}$ are uncorrelated. Covariance is bilinear, so

$$\mathrm{Cov}(\overline{X}, \overline{Y}) = \frac{1}{n^2}\sum_{i=1}^{n}\sum_{j=1}^{n} \mathrm{Cov}(X_i, Y_j).$$

For $i \neq j$, X_i depends only on P_i and Y_j only on P_j, which are independent, so $\mathrm{Cov}(X_i, Y_j) = 0$. For $i = j$, $\mathrm{Cov}(X_i, Y_i) = 0$ by part (i). Every term vanishes, so $\mathrm{Cov}(\overline{X}, \overline{Y}) = 0$: $\overline{X}$ and $\overline{Y}$ are uncorrelated.

The approximate probability. The X_i are independent and identically distributed with mean 0 and variance $\frac{1}{2}$. So

$$\mathrm{E}(\overline{X}) = 0, \qquad \mathrm{Var}(\overline{X}) = \frac{1}{n^2}\sum_{i=1}^{n} \mathrm{Var}(X_i) = \frac{1}{n^2} \cdot \frac{n}{2} = \frac{1}{2n},$$

where we may add the variances because the X_i are independent.

Ask yourself X is nothing like normal (look at its density). Why can we use a normal approximation?

The central limit theorem needs only independent, identically distributed terms with finite variance, whatever their shape. So for large n, $\overline{X}$ is approximately $\mathrm{N}(0, \frac{1}{2n})$, with standard deviation $\sigma = \dfrac{1}{\sqrt{2n}}$. Now

$$\sqrt{\frac{2}{n}} = \frac{2}{\sqrt{2n}} = 2\sigma,$$

so with $Z \sim \mathrm{N}(0,1)$,

$$\begin{aligned} \mathrm{P}\left(|\overline{X}| \leq \sqrt{\tfrac{2}{n}}\right) = \mathrm{P}\left(\frac{|\overline{X}|}{\sigma} \leq 2\right) &\approx \mathrm{P}(|Z| \leq 2) \\ &= 2\Phi(2) - 1 = 2(0.9772) - 1 \approx 0.954 \approx 0.95, \end{aligned}$$

as required.

Checks

- The variance $\frac{1}{2}$ makes sense: $X^2 + Y^2 = 1$ and, by symmetry, $\mathrm{E}(X^2) = \mathrm{E}(Y^2)$, so each must be $\frac{1}{2}$. No integration needed!
- A simulation of 4×10^5 random angles matched $F(x) = 1 - \frac{1}{\pi}\arccos x$ to about three decimal places at several values of x, and gave $\mathrm{Var}(X) \approx 0.5007$ and $\mathrm{Cov}(X, Y) \approx 0.001$.
- With $n = 50$ and 40 000 repetitions, the simulated $\mathrm{Var}(\overline{X})$ was 0.00997 (theory 0.01), and the proportion with $|\overline{X}| \leq \sqrt{2/50} = 0.2$ was 0.954, against $\mathrm{P}(|Z| \leq 2) = 0.9545$.

Common slips

- Differentiating $1 - \frac{1}{\pi}\arccos x$ with the wrong sign and getting a negative "density". A density can never be negative, so check the sign.

- Concluding independence from zero covariance. Independence implies zero covariance, but not the other way round.
- Using $\mathrm{Var}(\overline{X}) = \frac{1}{2}$ or $\frac{1}{2} \cdot \frac{1}{\sqrt{n}}$: the variance of a mean of n independent copies is the single variance divided by n; it is the *standard deviation* that scales like $1/\sqrt{n}$.

Beyond STEP

The density $1/(\pi\sqrt{1-x^2})$ is the *arcsine distribution*, which turns up surprisingly often, for example as the limiting law of the fraction of time a random walk spends positive. The rest of the question is about the gap between uncorrelated and independent, and about the central limit theorem, which you will prove using moment generating or characteristic functions in a first-year probability course. In fact $(\overline{X}, \overline{Y})$ is approximately bivariate normal, and for jointly normal variables zero correlation *does* imply independence. Grimmett and Welsh, *Probability: An Introduction*, covers all of this well.

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